

How to compile and install

Since we have seen that some users find the installation to be a challenge, here are the steps to extract and convert this tool into an executable file:

As discussed before, gpsim can be easily installed in Linux/Unix environment. However, gpsim is available in compressed format and should be converted to an executable format for installation:

1. Create a parent subdirectory for gpsim (example, gnupic)

```
$ mkdir ~/gnupic
```

(you can choose whatever directory name you like)

2. Copy the tar ball to this directory and expand it:

```
$ cd ~/gnupic
```

```
$ mv the_path_of_where_ever_the_tarball_is/gpsim-0.x.y.tar.gz.
```

```
$ tar -xvzf gpsim-0.x.y.tar.gz
```

(Where 'x.y' is the release number. As per the software in the DVD 'x.y' is '27.0')

These commands will create a subdirectory called gpsim-0.x.y. All of gpsim's source code will be untarred here.

3. `cd gpsim-0.x.y`

4. `./configure`

This will check your system for the proper tools necessary to build gpsim and then it will create Makefiles.

5. `make all`

This will create the executable.

6. `su root`

Su or root privileges are required to complete the installation. If you don't have root privileges or don't wish to install gpsim in the /usr/local/bin directory, check the steps on building without root privileges.

7. `make install`

This copies the executable file into the /usr/local/bin subdirectory.

The last and final step is to install gpsim:

1. Use the following codes:

```
cd $HOME/compiling
```

```
tar xzvf gpsim-VERSION-targz
```

```
cd gpsim-VERSION-
```

```
./configure --with-exdbm=$HOME/localinst/exdbm --prefix=$HOME/localinst/gpsim
```

```
make install
```

```
cd $HOME
```

2. Edit your shell configuration and add something like:

```
export PATH="$PATH:$HOME/localinst/gpsim/bin
```

3. Re-login to let changes take effect.

As these files were only needed for compiling and installation, they can be removed. To

remove these files you can use the commands:

```
rm -rf $HOME/compiling/exdbm
```

```
rm -rf $HOME/compiling/gpsim
```

```
rm -rf $HOME/compiling/gtk+extra
```

Use the following link to build and install gpsim on Windows: <http://gpsim.sourceforge.net/gpsimWin32/gpsimWin32.html>

cause of the bugs. gpsim helps diagnose the bugs providing that history.

Additional benefits through supportive tools

gpsim has the following supportive tools that help it in simulation:

Register reviewer. gpsim has two similar register windows that provide data for RAM and EEPROM. It allows you to see all registers in the current processor, with their display names and values. The register viewer provides the following functions:

1. One can not only set read, write and break point values but also clear each break point on the registers

2. Writes and reads of specific data can be logged in

3. Cells can be copied just by dragging the borders of the selected cells

4. The content of one cell can be copied to another cell

5. 'Add watch' allows one to watch the register values

Source browser. gpsim has two source browsers: .asm and .obj. The .asm is a colour-coded display of the PIC source whereas the .obj provides an opcode view. The opcode view has memory cell on one line and information related to disassembly, address and hexadecimal value on the other.

Breadboard. It not only allows one to create/modify and examine the PIC environment but also helps to instantiate PIC processor from the command line or from .stc file. In simple language, it helps in creating a correct configuration to test the source code. You can create code just by a click and also connect to those nodes for a bet-

THE COMPLETE MAGAZINE ON OPEN SOURCE

OpenSource
ForYou



Your favourite Magazine on Open Source is now on the Web, too.

OpenSourceForU.com

Follow us on Twitter @LinuxForYou